

ALI FETEHA

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AI Engineer · Azure AI Engineer Associate · IEEE Published Researcher

Professional Summary

Results-driven **AI Engineer** with 5+ years of combined experience designing and deploying production-grade intelligent systems. Specialized in **multi-agent architectures**, **Retrieval-Augmented Generation (RAG) pipelines**, **LLM fine-tuning**, and end-to-end **Azure AI** platform engineering. **Microsoft Certified Azure AI Engineer Associate** and **IEEE-published researcher** with a proven **96%+ Character Recognition Rate** on Arabic OCR. Demonstrated ability to translate R&D into scalable, business-ready AI products — from chatbots and intelligent search to agentic automation and enterprise data integration.

Core competencies: Generative AI · Agentic AI · LLMs · NLP · Computer Vision · Azure AI Services · RAG · Vector Databases · Semantic Kernel · FastAPI · Python · MLOps

Education

Egypt-Japan University of Science and Technology (EJUST)

Sep. 2020 – Feb. 2025

Bachelor of Science in Computer Science and Engineering · Grade: Very Good

Alexandria, Egypt

- Graduation Thesis: Arabic Handwritten Document OCR — achieved 96%+ CRR; published at **IEEE NILES 2024**

Professional Experience

Applab

Apr 2025 – Present

AI Engineer

Doha, Qatar

- **Architected and deployed production AI solutions** on Azure (Azure AI Services, Container Apps, Azure AI Search, Azure Functions, Speech Services, Azure Storage), enabling secure, scalable intelligent platforms serving enterprise users.
- **Designed multi-agent architectures** using Semantic Kernel and agentic AI frameworks, automating complex multi-step enterprise workflows and reducing manual processing time significantly.
- **Built RAG pipelines and AI-powered chatbots** with intelligent semantic search, improving user engagement and enabling accurate, context-aware responses over large enterprise knowledge bases.
- **Engineered data pipelines** for structured and unstructured document extraction, web scraping, and knowledge graph construction — powering downstream AI reasoning and retrieval systems.
- **Implemented enterprise SharePoint integration** using SharePoint SDK and Microsoft Graph API, delivering secure, automated, and scalable data ingestion workflows for AI platforms.
- **Led AI proof-of-concept development** from requirements gathering through delivery, aligning technical AI capabilities with business objectives and demonstrating ROI to stakeholders.
- Leveraged **Copilot Studio** and **Power Automate** to extend AI solutions into Microsoft 365 ecosystems, enabling non-technical teams to interact with intelligent automation workflows.

Udacity

Sep 2022 – Dec 2022

Advanced Android Development Nano-degree Graduate

- Delivered multiple production-quality **Android applications in Kotlin** using MVVM architecture, ViewModels, LiveData, Room, Retrofit, and Glide; implemented coroutines for async processing and custom UI animations.

Hassan Aly Hassan Association for Contracting

May 2020 – Feb 2025

Accountant & Technical Office Engineer

Alexandria, Egypt

- Managed end-to-end financial operations (payroll, budgeting, asset tracking, bank reconciliation) and coordinated engineering quantity analysis across multiple active construction projects.

Projects

AHDoc — Arabic Handwritten Document OCR | Python · Deep Learning · Swin v2 · BERT · ResNet-50 Sep 2024 –

End-to-end OCR system converting Arabic handwritten documents into structured CSV output · IEEE Published

- Achieved **96%+ Character Recognition Rate (CRR)** using a **Swin Transformer v2 – BERT** vision encoder-decoder architecture trained with custom data augmentation and transfer learning strategies.
- Fine-tuned a word detection model using **Differentiable Binarization + Adaptive Scale Fusion** (ResNet-50 backbone), reaching **81.66% precision** and **79.0% F-measure** on Arabic line segmentation — competitive with state-of-the-art benchmarks.
- Engineered automated **data generation and augmentation pipelines**; applied transfer learning to overcome OCR sequence-length constraints during training.
- Research published at **IEEE NILES 2024** conference (DOI: 10.1109/niles63360.2024.10753216).

MBTI Personality Classifier <i>Python · NLP · BERT · LSTM · CBOW · Scikit-learn</i>	Nov 2023 – Jun 2024
<i>NLP classification system predicting 16 MBTI personality types from social text data</i>	
Benchmarked three architectures end-to-end: LSTM (73%), CBOW (76%), and fine-tuned BERT (80% accuracy) — demonstrating strong command of transformer fine-tuning and NLP evaluation methodology.	
ImgVox — 2D to 3D Voxel Reconstruction <i>Python · PyTorch · PyTorch Mobile · 3D AutoEncoder</i>	Feb 2023 – Apr 2023
<i>On-device 3D reconstruction from 2D images deployed via PyTorch Mobile</i>	
Trained a 3D AutoEncoder for voxel grid reconstruction; optimized and exported model as .ptl files using PyTorch Mobile for local inference on mobile devices — demonstrating full ML deployment lifecycle.	
NetworkGraphI — Twitter Network Analysis <i>Java · Selenium · JUNG · SQLite · Multi-threading</i>	Mar 2023 – Apr 2023
<i>Large-scale social network analysis across 500K+ Twitter accounts</i>	
Built a multi-threaded Selenium scraper in Java to crawl 500K+ account connections; applied graph analysis algorithms (JUNG) to reveal structural patterns and community relationships in the network.	
Grocery Shopping App <i>Kotlin · Android · MVVM · PostgreSQL · SQLite</i>	Oct 2023 – Dec 2023
<i>Full-stack Android e-commerce application with remote/local caching strategy</i>	
Built from scratch with MVVM + coroutines, implementing advanced authentication, secure key management, and a dual remote/local caching architecture for responsive, offline-capable UX.	

Technical Skills

- AI & Machine Learning:** Generative AI, Agentic AI, Multi-Agent Systems, Large Language Models (LLMs), RAG Pipelines, LLM Fine-tuning & Evaluation, NLP, Computer Vision, OCR, Deep Learning, Transfer Learning, MLOps
- Frameworks & Libraries:** Semantic Kernel, LangChain, PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, FastAPI, Hugging Face Transformers
- Azure & Cloud:** Azure AI Services, Azure AI Search, Azure Container Apps, Azure Functions, Azure App Services, Azure Speech Services, Azure Storage, Microsoft Graph API, SharePoint SDK, Copilot Studio, Power Automate
- Programming Languages:** Python (advanced), C++, Java, Kotlin, SQL
- Engineering Practices:** REST API design, Data Pipelines, Web Scraping, DevOps, MVVM / MVC / MVP, Distributed Systems, Database Design, Git, Unix/Linux

Publications

Waly, A., Tarek, B., **Feteha, A.**, Yehia, R., Amr, G., & Fares, A. (2024). Arabic handwritten document OCR solution with binarization and adaptive scale fusion detection. *2024 6th Novel Intelligent and Leading Emerging Sciences Conference (NILES)*, pp. 316–319. doi.org/10.1109/niles63360.2024.10753216

Certifications

Microsoft Certified: Azure AI Engineer Associate (AI-102)	Sep 2025
<i>Microsoft — Azure AI solution design, Cognitive Services, Azure AI Search, Bot Service</i>	
Retrieval Augmented Generation (RAG) with LangChain	Feb 2025
<i>DataCamp — RAG architecture, vector retrieval, LangChain orchestration</i>	
ChatGPT Prompt Engineering for Developers	Sep 2024
<i>DataCamp — Prompt engineering patterns, chain-of-thought, few-shot techniques</i>	
Developing Large Language Models	Aug 2024
<i>DataCamp — LLM training, LLM Ops, 4 courses + 1 project</i>	
Deep Learning in Python	Aug 2024
<i>DataCamp — PyTorch, neural networks, 4 courses + 1 project</i>	
Advanced Android Development Nano-Degree	Sep – Dec 2022
<i>Udacity — Kotlin, Android Architecture, Agile, 7 chapters + projects</i>	